

Kendall Ash

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EDUCATION

Michigan State University MS Fisheries & Wildlife; EEB	East Lansing, MI Expected December 2026
University of Michigan BS, <i>summa cum laude</i> , Ecology & Evolutionary Biology; German	Ann Arbor, MI May 2024

FELLOWSHIPS AND AWARDS

Summer Fellowship in Ecology, Evolution, and Behavior, EEB, MSU	2026
NSF GRFP – Honorable Mention	2026
MSU Bug House Fellowship	2026
John and Marietta Peter Scholarship, College of Agriculture and Natural Resources, MSU	2025-2026
EEB Symposium Art In Science Runner Up	2025
Robert C. Ball and Betty A. Ball Fisheries and Wildlife Fellowship, MSU	2025
International Conference on Harmful Algae Travel Award, NOAA NCCOS	2025
Fisheries and Wildlife GSO Travel Award, MSU	2025
EEB International Travel Award, MSU	2025
Best Prospective Oral Presentation, Fisheries and Wildlife GSO Symposium, MSU	2025
James B. Angell Scholar, University of Michigan	2022 & 2024
University Honors, University of Michigan	2020-2024

RESEARCH EXPERIENCE

Michigan State University <i>Graduate Research Assistant in Bridge Lab</i> https://bridgelabresearch.org/	East Lansing, MI January 2025 – December 2026
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Conducting large-scale data analysis on U.S. lakes using the NLA, LAGOS, and CMIP6 databases

- Using machine learning to analyze the drivers of cyanobacteria microcystin production across thousands of lakes in the present day and under future climate change scenarios
- Analyzing cyanobacteria community composition and functional traits using redundancy analysis
- Compiling, processing, and QA/QC'd large amounts of data using RStudio and cloud computing
- Assisting with additional lab projects including analyzing the relationship between lake sampling efforts and community demographics as well as creating the LAGOS LIMNO v.2.0 database

University of Michigan Biological Station <i>Field Technician</i>	Pellston, MI May 2024 – August 2024
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Assisted visiting researchers with projects at the station

- Assembled mesocosm plot for ant colony research with Dr. J. P. Lessard (Concordia University)
- Captured bumble bees for population and pathogen surveys and collected strawberries for chemical analysis with Dr. Mary Jamieson (Oakland University)
- Digitized data archive and herbarium specimens
- Mapped field sites on station property on ArcGIS

University of Michigan <i>Research Assistant with Dr. Meghan Duffy</i>	Ann Arbor, MI June 2023 – May 2024
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Examined relationships and complex interactions between fungal and microsporidian parasites of *Daphnia*

<https://duffylab.wordpress.com/>

- Field work: sampled and analyzed local lakes using YSI multiparameter probe, fluorometer, Secchi disk, and zooplankton nets

- Lab work: assisted with and conducted various research projects, maintained *Daphnia* and parasite colonies, filtered lake water, ran genetic analyses

Michigan Medicine

Ann Arbor, MI

Research Assistant with Dr. Rosanne Rouf

August 2021 – January 2022

Used mouse models to examine sex differences in heart failure and cardiovascular disease

- Conducted genetic analysis to identify mice genotypes using DNA extraction, PCR, and gel electrophoresis

PERSONAL PROJECTS

***Daphnia* feeding assay (Duffy Lab)**

August 2023 – November 2023

Conducted algal feeding assay to compare efficacy of cell quantification methods to assess *Daphnia* feeding

- Quantified algal samples before and after *Daphnia* feeding using ddPCR with and without addition of PMA dye, fluorometry, and cell counting
- Conducted statistical analysis and visualization in RStudio and wrote a comprehensive report assessing efficacy of quantification methods

Arctic algal growth experiment (SIT Study Abroad Program)

April 2023 – May 2023

Ran a temperature-dependent algal growth experiment to project climate change effects on harmful algal blooms in Iceland. Conducted under supervision of Sara Hardardottir at the Marine and Freshwater Research Institute in Hafnarfjörður, Iceland.

- Collected, cultured, and counted *Pseudo-nitzschia* diatoms
- Ran statistical analysis in RStudio, wrote and presented a comprehensive report

PUBLICATIONS, PRESENTATIONS, AND POSTERS

Lippert, A.L., Z. Naylor, X. Sun, P.J. Hanly, J. Díaz Vázquez, M. Haite, **K. Ash**, K.S. Cheruvilil, and P.A. Soranno. 2025. User Guide to LAGOS-US HUMAN v1: Data module of human population(1990-2020), urbanization classification, and lake access in the conterminous U.S. ver 1. Environmental Data Initiative.

K. Ash. Toxic relationships: predicting drivers of lake microcystin across the U.S. and into the future. Aquatic Confluence: Science, People, Knowledge (ASLO-SIL 2026 Joint Meeting). May 2026.

K. Ash. Predicting cyanobacteria blooms in U.S. lakes using national scale data. Oral presentation, International Conference on Harmful Algae, Punta Arenas, Chile. October 2025.

A. Lippert, **K. Ash**. Examining relationships between lake sampling efforts and community demographics. Poster presentation, MSU Ecology, Evolution, and Behavior Symposium. May 2025

K. Ash. Predicting occurrences of cyanobacteria harmful algal blooms in U.S. lakes under a changing climate. Oral presentation, MSU Fisheries & Wildlife GSO Symposium. March 2025.

TEACHING EXPERIENCE

Michigan State University

Magaliesburg and Sodwana Bay, South Africa

Teaching Assistant, Undergraduate Internship Experiences in South Africa

2025

- Coordinated and supported students for two-week cheetah conservation and marine biology internships in South Africa; graded student assignments

Michigan State University

East Lansing, MI

Grader, Field and Laboratory Techniques for Aquatic Studies

2025

- Prepared and assisted with laboratory activities and equipment, graded student assignments
- Prepared and presented a guest lecture about algae and harmful algal blooms

SKILLS

- Programming in R, Python, & C++
- Machine learning (LightGBM, XGBoost, RandomForest)
- Microsoft Office Suite and Adobe Creative Cloud
- RStudio, ArcGIS, QGIS, Google Earth Engine, and Google Earth Pro
- Code and data sharing through GitHub, Zenodo, EDI
- Climate modeling and remote sensing

- Statistical analyses: RDA , PCA, GLM
- Cloud Computing (AWS, MSU HPCC)
- Aquatic field techniques—zooplankton and phytoplankton collection, YSI probes
- Insect capture and collection
- Microscopy, DNA extraction, PCR, ddPCR, gel electrophoresis, cell counting, fluorometry
- Mass spectrometry, chromatography
- Open Water SCUBA Certified
- Proficient in German

AREAS OF INTEREST

Aquatic ecology	Climate change and Arctic ecology	Microbial ecology	Open Science
Computational ecology	Harmful algal blooms	Inducible defense	Parasitism
			Biomagnification

EXTRACURRICULARS

Organizer, iNaturalist City Nature Challenge Lansing	2025, 2026
iNaturalist ambassador	2026
Guide at MSU Bug House	2025
Volunteer at Refugee Development Center	2025
Tutor in biology, chemistry, math, and German	2016-2024
University of Michigan Women's Rugby	2023-2024
University of Michigan Entomology Club	2022-2024
Volunteer at Ann Arbor Parks and Recreation & Campus Farm	2021-2024

RELEVANT COURSEWORK

Introductory and organic chemistry	Protecting & restoring lakes
General physics I & II	Evolution
Calculus I, II, & III	Climate modeling & carbon management
Introductory biology	Arctic: changing ecosystems and resilience
General ecology	Fisheries & Wildlife Outreach and Engagement
Genetics	Applications of Terrestrial Remote Sensing
Biology of insects	Statistical Methods in Ecology and Evolution I & II
Plant biology	Molecular Ecology & Conservation Genetics

REFERENCES

Dr. Kendra Spence	Dr. Patrick Hanly	Dr. Helen Habicht	Daniel Govoni
Cheruvelil	Advisor	Research coordinator	Study abroad
Advisor	Project Scientist	University of	professor/ project
PI/Co-director	Bridge Lab MSU	Michigan	mentor
Bridge Lab MSU	hanlypat@msu.edu	Biological Station	School for
ksc@msu.edu		habicht@umich.edu	International Training
			danielgo@vedur.is